Wellington Bristow

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Education

University of Southern California — Los Angeles, CA
Doctor of Philosophy, Aerospace Engineering (GPA: 4.00) — *Aug. 2026 – May 2030*

University of Southern California — Los Angeles, CA
Bachelor of Science, Aerospace Engineering (GPA: 3.78) — *Jan. 2023 – May 2026*

Experience

Graduate Research Assistant — Aug. 2026 – Present
Combustion and Fuels Research Laboratory, USC (Advisor: Dr. Egolfopoulos) — *Los Angeles, CA*
- Improved experimental electonic and mechanical systems to reach pressures of interest to IC engines
- Identified limitations in PeleLMeX then collaborated with Dr. Marc Day to facilitate fixes for 3D simulations
- Connected legacy Fortran codes with user-friendly shell scripts to enable use within research workflow

Undergraduate Research Assistant — Oct. 2024 – Aug. 2026
Combustion and Fuels Research Labratory, USC (Advisor: Dr. Egolfopoulos) — *Los Angeles, CA*
- Led a novel effort to produce reliable unstable flame speeds conditions of interest to IC engines
- Produced strong experimental evidence in support of the unproven theory of "self-turbulization"
- Implemented large scale PeleLMeX direct numerical simulations on GPU centered HPC hardware
- Wrote Python and MATLAB scripts to identify mixtures and process data for 25+ initial mixtures
- Set up a single flow reactor for PIV imaging to capture transient pulsed plasma-induced fluid turbulence
- Gave 4 presentations of experimental and numerical findings to Dr. Poledunenko's group at UCONN
- Presented 3 posters at the 41st International Symposium on Combustion regarding H_2 and gasoline fuels
- Led a 4 person senior project on the effect of pressure in "self-turbulization" (unpublished report & poster)

Biomedical Researcher — May 2026 – Aug. 2026
Universal Transplant Company — *Los Angeles, CA*
- Identified connection between dispersion ultrafiltration system modeling and patented antibody removal device
- Adapted models to identify dynamic conditions which create favorable conditions for antibody removal
- Conducted 80+ paper literature review for SBIR NIH grant application on ABOi desensitization techniques
- Drafted initial version of an SBIR NIH grant for potential Phase I + Phase II funding

Course Producer AME 341 a&b — Aug. 2025 – May 2026
Department of Aerospace and Mechanical Engineering, USC — *Los Angeles, CA*
- Instructed 120+ students during fast-paced electronic and engineering experiments through 3 lab sections
- Mentored 16 students' 4 unique junior projects on propeller performance and heat transfer experiments
- Provided feedback on course material and technical writing for +130 assignments and +30 presentations

Manufacturing Engineer — Jan. 2023 – Aug. 2025
SC Racing Formula SAE Internal Combustion Engine Team, USC — *Los Angeles, CA*
- Only student to program and machine 4 uprights, 160+ hours each, mentored successive student machinists
- Manufactured components for aerodynamics, suspension and chassis subteams via TIG welding and CNC
- Instructed 8 students on TIG welding fundamentals and techniques relevant to chassis manufacturing
- Performed detailed work under tight deadlines to help produce one formula style car per school year

Student Machinist — Aug. 2023 – Aug. 2025
Viterbi School of Engineering, USC — *Los Angeles, CA*
- Built fundamentals manual & CNC lathe and mill competence, mentored next group of student machinists
- Learned advanced indication techniques to produce the most technical design team components at USC

Volunteer Varsity Quarterbacks Coach — Aug. 2022 – Nov. 2022
Taft Charter High School, Varsity Football Team — *Woodland Hills, CA*
- Mentored group of 4 first time QBs, taught fundemental mechanics and understanding of the position
- Collaborated with head coach on game preparation along with play-calling in high pressure situations

Sales Associate — Aug. 2022 – Jan. 2023
Play It Again Sports, Sherman Oaks — *Sherman Oaks, CA*
- Worked for a retail store selling & buying used sports equipment to/from customers
- Taught myself baseball glove repair and breaking-in practices to service customers outside of work

Publications

4. Wellington Bristow, Kyuho Van, Sai Sandeep Dammati, Alexei Poludenko, and Fokion N. Egolfopoulos. Experimentally exploring self-turbulization of spherically expanding lean hydrogen flames. *Combustion and Flame*, 2027 [*Work in Progress*]

3. Kyuho Van, Suhyeon Oh, Wellington Bristow, Ji-Woong Park, Yuanjiang Pei, and Fokion N. Egolfopoulos. A numerical investigation of hydrogen flame instabilities at high pressures and temperatures. *Combustion and Flame*, 2026 [*Pending Submission*]

2. Suhyeon Oh, Kyuho Van, Anguo Hu, Wellington Bristow, Ji-Woong Park, Yuanjiang Pei, and Fokion N. Egolfopoulos. Propagation of laminar hydrogen flames at engine conditions. *Combustion and Flame*, 2026 [*Pending Submission*]

1. Suhyeon Oh, Kyuho Van, Wellington Bristow, Ji-Woong Park, Yuanjiang Pei, and Fokion N. Egolfopoulos. Propagation of laminar flames of gasoline-type fuels at engine conditions. *Combustion and Flame*, 2026 [*Pending Submission*]

Presentations

4. Wellington Bristow, Kyuho Van, Suhyeon Oh, Ji-Woong Park, Yuanjiang Pei, and Fokion N. Egolfopoulos. A numerical investigation of hydrogen flame instabilities at high pressures and temperatures. 41st International Symposium on Combustion, July 2026. Poster ID: 3-02-03, Organized by The Combustion Institute

3. Wellington Bristow, Suhyeon Oh, Kyuho Van, Anguo Hu, Ji-Woong Park, Yuanjiang Pei, and Fokion N. Egolfopoulos. Propagation of laminar hydrogen flames at engine conditions. 41st International Symposium on Combustion, July 2026. Poster ID: 2-02-01, Organized by The Combustion Institute

2. Wellington Bristow, Suhyeon Oh, Kyuho Van, Ji-Woong Park, Yuanjiang Pei, and Fokion N. Egolfopoulos. Propagation of laminar flames of gasoline-type fuels at engine conditions. 41st International Symposium on Combustion, July 2026. Poster ID: 1-02-06, Organized by The Combustion Institute

1. Wellington Bristow, Eli Goldenberg, Rachel McCafferty, and Grace Kanaley. Experimentally exploring self-turbulization of spherically expanding lean hydrogen flames. Senior Projects Exposition, December 2025. Group 24, Department of Aerospace and Mechanical Engineering, USC